

Supplementary Material S4

Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements Table 17 in the manuscript (11 studies of greatest translational traceability, Section 6.3). Section A presents the 23 eligible studies organised by clinical domain (D1–D4); Section B presents the 7 contextual studies outside the eligible set.

The internal # column corresponds to the corpus order shared with Supplementary Material S2, not to bibliographic reference numbers. **#18 corresponds to Ahmad et al. (2024), retracted after corpus finalisation (17 August 2026, *Healthc Technol Lett* 2026;13(1):e70095); excluded with retroactive effect. The gap at #18 is not a numbering error.** The 4 retracted studies do not appear in this table.

Section A. Eligible corpus (23 studies)

#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
<i>Domain D1: Automated facial expression (8 studies)</i>							
1	Johnson et al. (2025)	Emotion recogn.	AffectNet + clinical	ASD children (N=48) / Lab./Ecol.	EfficientNet-v2	ACC 67.8%	Clinical
8	Radočaj & Mart. (2025)	Emotion recogn.	FERAC	ASD children / Lab.	Swin Transformer	ACC 80%, F1 78.9%	Partial prov.
9	Washington et al. (2022)	Emotion recogn.	CAFE Sub. A + clinical	Children / Lab./Ecol.	ResNet-152	Bal.ACC 79.1%, F1 78%	Clinical
10	Sholikah et al. (2024)	Real-time recogn.	FER-2013	N=3 ASD (test) ^a / Mobile	VGG-16	ACC 91%, F1 91%	Public gen.
21	Aljbori et al. (2025)	Emotion recogn.	Kaggle ASD	ASD children / Lab.	3D-CNN + LSTM	ACC 99.82%, F1 98.6%	Web
23	Vedantham et al. (2025)	Emotion recogn.	Kaggle	Children / Lab.	GWO-GEfficientNet	ACC 96.5%, F1 96.4%	Web
27	Dapogny et al. (2019)	Facial expr.	JEMImE (N=157)	Children 6–11 / Clin. game	Rand. Forests + PCRF	ACC ~82%	Public cur.

(Continues on next page)

(Continued from previous page)

#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
28	Leo et al. (2018)	Facial prod. ASD	CK++clinical (N=17)	ASD children / Clinical	HOG + SVM + CNN	TP 85% (CK+) ^b	Clinical
<i>Domain D2: Affect and therapeutic engagement (5 studies)</i>							
4	Lecciso et al. (2021)	Therapeutic interv.	Private (N=24)	ASD children / Therapy	OpenFace	g=0.52	Clinical
13	Hassan et al. (2025)	Therapy monit.	DREAM (N=61)	ASD children 3–6 / Robot th.	RF + XGBoost + ExtraT.	ACC 95.6%, F1 92.8%	Public cur.
15	Rudovic et al. (2018)	Affect / therapy	MIT private (N=35)	ASD children / Robot th.	PPA-net	ICC ~60%	Clinical
16	Cicceri et al. (2026)	Pointing gestures	Private clinical	ASC children / Clinical	4-level framework	mAP 88.6%	Clinical
17	Rudovic et al. (2019)	Active learning	MIT private (N=35)	ASD children / Robot th.	4 LSTMs + Deep RL	Fig. ^c	Clinical
<i>Domain D3: Emotion recognition intervention (3 studies)</i>							
2	He et al. (2019)	Eye-tracking	Private (N=42)	ASD preschoolers n=21 / Clinical	Eye-tracking	r=0.619	Clinical
3	Reed et al. (2021)	Epidemiological	ALSPAC (N=3,562)	Paediatric cohort / Population	Regression	b=−0.18	Cohort
18	Almeida et al. (2019)	Serious game	No ML dataset	ASD 6–12 / Game	RPG (no ML)	ISO/IEC usab.	N/A
<i>Domain D4: Screening or diagnosis of ASD (7 studies)</i>							
5	Shaik et al. (2025)	ASD detection	Kaggle ASD	Children / Lab.	Ensemble CNN	ACC 84%, F1 81.9%	Web
12	Sariyanidi et al. (2025)	Autism pred.	CHOP clinical (N=42)	ASD + TD / Clinical	Dict. Learn. + SVM	Fig. ^c	Clinical
14	Alsubai (2025)	ASD detect. (tab.)	Kaggle screening	12–36 months / Screening	Transformer MDNCT	ACC 94.7%	Web
19	Alam et al. (2025)	ASD detection	Kaggle + UIFID	Children / Lab.	ASD-UANet	ACC 96%	Web
20	Hwang et al. (2024)	DD screening	Private (N=89)	34–77 months / Mobile	LSTM + 2D-FAN	ACC 88%, F1 80%	Clinical

(Continues on next page)

(Continued from previous page)

#	Study	Clinical task	Dataset	Population / Setting	Architecture	Metrics	Data prov.
24	Hosney et al. (2024)	ASD detection	Kaggle (co-auth.) ^d	Children / Lab.	YOLOv8 + DCNN	ACC 97.2%, F1 93%	Web
29	Akın-Bülbül et al. (2024)	ET screening	Private ET	Infants 18–36m / Clinical	KNN + features	ACC 81.45%	Clinical

Section B. Contextual evidence outside the eligible set (7 studies)

Four infringe CE(3) (exclusively non-facial modalities) and three infringe the age criterion (exclusively adult participants).

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.	Reason
C1	El Rhatassi et al. (2024)	Emotion recogn.	FER-2013 + Kaggle autism	Autistic adults	Lab.	CNN vs ViT	ACC 65–74%	Public gen.	Adult participants
C2	El Rhatassi et al. (2025)	Real-time recogn.	CK++ + HeyGen AI	Autistic adults	Lab.	CNN×3 + LSTM + CC	ACC val. 53%	Mixed	Adult participants
C3	Levy et al. (2024)	Wearable system	FER-2013	ASD adults (N=10)	Ecol.	CNN + HW wearable	Usability	Public gen.	Adult participants
C4	Nagae & Lee (2022)	Therapy + EDA	Private EDA (N=11)	DD children	Robot th.	EDA + wavelet + SVM	Conf. matrix	Clinical	EDA only (CE3)
C5	Papaioannou et al. (2021)	EEG cogn. load	EEG (N=63)	ASD + ADHD adults	Clinical	MSE + PLSC	ANOVA	Clinical	EEG only (CE3)
C6	Xu & Ju (2023)	VBM brain	ABIDE (N=203)	ASD 8–12 (n=98)	Clinical	VBM	GMV+ADOS corr.	Public cur.	sMRI only (CE3)
C7	O'Sullivan et al. (2023)	Speaker diariz.	Clin. trial audio (N=72)	ASD 5–45	Home	Pyannote + xvector	F1 76.5%	Clinical	Speech only (CE3)

Notes.

- a. 3 ASD participants in Sholikah et al. (2024) involved in a usability test, not in metric computation.

- b. TP 85% for Leo et al. (2018) corresponds to the CK+ public dataset of typically developing adults, not to the 17 ASD children.
 - c. Fig.: metrics reported in graphical format only, without numerically extractable values. Applies to Rudovic et al. (2019) and Sariyanidi et al. (2025).
 - d. (co-auth.): Kaggle dataset created by a co-author of Hosney et al. (2024), constituting a documented conflict of interest.
1. Internal # column uses corpus order (shared with S2) for eligible studies, C-prefix for contextual. Does not correspond to bibliographic reference numbers. #18 is absent (Ahmad, retracted); this is not a numbering error.
 2. Retracted studies (Güngör et al., 2025; Mahmood et al., 2025; Kumar et al., 2024; Ahmad et al., 2024) do not appear. Ahmad et al. (2024) retracted 17 August 2026 (*Healthc Technol Lett* 2026;13(1):e70095); all four documented in Section 2.7 of the manuscript.
 3. Bal.ACC for Washington et al. (2022): balanced accuracy 79.1% on CAFE subset with human agreement $\geq 60\%$; full CAFE set value is 66.9%.
 4. Almeida et al. (2019): systems design without ML classification; ISO/IEC usability metrics are not comparable with supervised classification metrics.
 5. Contextual studies in Section B were appraised with MMAT in Supplementary Material S2 (Section B) but do not participate in the aggregate MMAT distribution of the manuscript.